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ORIGINAL ARTICLE

Hybrid CNN + LSTM + SVM model for advanced brain tumor classification

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Abstract

Cancer remains the second leading cause of death worldwide, with brain tumors contributing significantly to cancer-related fatalities. Early and accurate diagnosis of brain tumors is crucial for timely therapeutic intervention and improved patient outcomes. Recent advancements in medical imaging, particularly in image classification, have notably enhanced the efficacy of computer-aided diagnostic systems. This paper proposes five hybrid models: VGG16+LSTM+SVM, VGG19+LSTM+SVM, ResNet50+LSTM+SVM, ResNet152+LSTM+SVM, and EfficientNetB3+LSTM+SVM, for the multi-class classification of brain tumors. These models combine the feature extraction power of convolutional neural networks (VGG16, VGG19, ResNet50, ResNet152, and EfficientNetB3) with the sequential learning capabilities of Long Short-Term Memory (LSTM) networks and the robust classification strength of Support Vector Machines (SVM). The methodology begins with data augmentation to address class imbalance and increase the diversity of the training dataset. Features extracted by the convolutional layers are fed into LSTM networks, which capture temporal dependencies, and the resulting outputs are classified using SVM. Experimental results demonstrate that the proposed models outperform current state-of-the-art approaches, with the ResNet152+LSTM+SVM model achieving 99% accuracy, 99% precision, 99% recall, and 99% F1 score using the RMSprop optimizer.

Keywords Brain Tumor · MRI · Classification · VGG16 · VGG19 · ResNet50 · ResNet152 · EfficientNetB3 · LSTM · SVM

1 Introduction

Brain tumors represent one of the most serious health threats worldwide, affecting millions of individuals each year and contributing significantly to global mortality and morbidity. The impact of brain tumors extends beyond the medical domain, influencing patient quality of life, healthcare systems, and socioeconomic structures. In recognition of their severity and the need for continuous awareness, June 8th has been designated as World Brain Tumor Day since 2000. A brain tumor arises from the uncontrolled proliferation of abnormal cells within or around the brain, an organ composed of approximately 100 billion neurons responsible for regulating activities essential for human survival (Louis et al. 2016). Owing to the brain's structural and functional complexity, even small abnormalities can lead to life-threatening conditions. Brain tumors continue to be among the leading causes of morbidity and mortality in both adults and children in developed countries (Petruzzi et al. 2013). Based on biological behavior, tumors are broadly classified as benign or malignant. Benign tumors remain localized and often amenable to surgical intervention, whereas malignant tumors are composed of highly invasive cancerous cells capable of spreading to adjacent brain structures and the spinal cord, resulting in severe clinical consequences (Ayadi et al. 2021; Asif et al. 2023a).

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Among the various tumor types, gliomas, meningiomas, and pituitary tumors are the most frequently encountered. Gliomas, originating from glial cells, account for a large proportion of primary malignant brain tumors and exhibit the highest mortality rates (DeAngelis 2001). They are typically categorized into low-grade and high-grade gliomas, with high-grade gliomas being particularly aggressive and associated with a median survival time of approximately two years. Meningiomas develop from the meninges, the protective membranes surrounding the brain and spinal cord, and although often benign, can lead to neurological impairments due to compression of critical brain structures (Louis et al. 2016). Pituitary tumors arise from the pituitary gland, which plays a key role in endocrine regulation. While most pituitary tumors are benign, they may cause irreversible hormonal disturbances or vision loss if left untreated (Komninos et al. 2004). Given the wide heterogeneity in tumor characteristics such as location, shape, size, and internal tissue composition accurate and early diagnosis is essential to guide effective therapeutic intervention and improve patient prognosis.

Magnetic Resonance Imaging (MRI) remains the primary non-invasive imaging modality for the detection, characterization, and monitoring of brain tumors. Compared to X-ray or CT imaging, MRI provides superior soft-tissue contrast and avoids exposure to ionizing radiation. Common MRI sequences include T1-weighted (T1W) and T2-weighted (T2W) scans: T1W scans offer high structural detail, while T2W scans enable clear differentiation between normal and abnormal tissue regions (Rezaeijo et al. 2023). Despite its advantages, MRI interpretation is dependent on the expertise of radiologists, who must manually analyze numerous images. This process is time-consuming, prone to inter-observer variability, and may lead to inconsistencies in diagnosis—especially in early-stage tumors or cases with subtle visual characteristics. These challenges have increased demand for computer-aided diagnosis (CAD) systems, which support radiologists by enabling automated, objective, and reproducible tumor analysis (Amin et al. 2020).

Machine Learning (ML) and Deep Learning (DL) techniques have shown great promise for automating the classification of brain tumors in MRI images. Early approaches relied on handcrafted features, classical ML classifiers, and statistical analyses. More recently, DL-based methods—particularly Convolutional Neural Networks (CNNs)—have demonstrated superior performance by automatically learning discriminative features from imaging data (Qiu et al. 2017). Approaches such as transfer learning (TL) (Swati et al. 2019), deep neural architectures (Muhammad et al. 2020), fusion-based feature representations (Saba et al. 2020), and classical ML algorithms (Zhou et al. 2018; Lavanyadevi et al. 2017) have been extensively explored. DL methods, in particular, have demonstrated superior capability in addressing complex classification tasks compared to traditional ML approaches (Qiu et al. 2017). With continuous advancements in network design, numerous studies have successfully employed Convolutional Neural Networks (CNNs) for robust and accurate brain tumor diagnosis, further validating their effectiveness in medical imaging applications (Toğaçar et al. 2020; Deepak and Ameer 2019).

Despite substantial progress, several challenges remain in achieving reliable multi-class brain tumor classification. MRI images exhibit high variability in tumor size, shape, intensity distribution, and anatomical location. Traditional CNNs primarily capture local or hierarchical patterns but may not effectively model global or sequential spatial dependencies across tumor regions. Furthermore, end-to-end CNN-based classification models sometimes fail to generalize well across heterogeneous datasets. Additionally, many existing methods rely solely on softmax classifiers, which may be insufficient for separating complex, high-dimensional feature spaces. These limitations highlight the need for a computationally efficient, fully automated, and robust classification framework that integrates multiple learning components to address different aspects of feature extraction and decision-making.

To address these challenges, this paper introduces a set of hybrid deep learning and machine learning models that leverage the complementary strengths of multiple architectures. Specifically, convolutional neural networks (CNNs), including VGG16, VGG19, ResNet50, ResNet152, and EfficientNetB3, are used to perform hierarchical and discriminative feature extraction from MRI images. The extracted multi-level feature maps are then processed by long short-term memory (LSTM) networks, which capture spatial and long-range dependencies that conventional CNNs may overlook. Finally, support vector machines (SVMs) are applied for robust multi-class classification in high-dimensional feature spaces, benefiting from their ability to construct optimal decision boundaries even in complex, nonlinear domains.

By integrating CNNs, LSTMs, and SVMs into a single framework, the proposed approach combines the representation power of deep learning with the discriminative strength of traditional machine learning. This hybrid integration improves classification accuracy, generalization, and robustness to variations in MRI data. Moreover, the pipeline is fully automated, reducing the need for manual intervention and enhancing clinical workflow efficiency, making it suitable for real-world computer-aided diagnosis (CAD) applications. The major contributions of this work are summarized as follows:

- A novel set of hybrid architectures combining CNNs (VGG16, VGG19, ResNet50, ResNet152, EfficientNetB3), LSTM networks, and SVM classifiers for enhanced multi-class brain tumor classification.
- A sequential modeling mechanism in which LSTM networks capture long-range spatial relationships across CNN-extracted feature maps.
- A rigorous comparative evaluation of five hybrid models—VGG16+LSTM+SVM, VGG19+LSTM+SVM, ResNet50+LSTM+SVM, ResNet152+LSTM+SVM, and EfficientNetB3+LSTM+SVM—on a unified dataset to assess performance differences.
- A fully automated classification framework requiring minimal manual preprocessing and designed for seamless integration into computer-aided diagnosis (CAD) systems.
- Extensive experimental results demonstrating that the proposed hybrid models outperform several existing approaches in terms of accuracy, precision, sensitivity, and overall robustness.

The structure of this paper is organized as follows: Sect. 2 reviews relevant studies, providing an in-depth overview of previous work in the field. Section 3 presents a detailed description of the proposed VGG16+LSTM+SVM, VGG19+LSTM+SVM, ResNet50+LSTM+SVM, ResNet152+LSTM+SVM, and EfficientNetB3+LSTM+SVM models, emphasizing their design and integration. Section 4 reports the experimental results and provides a discussion of the findings. Finally, Sect. 5 concludes the paper by summarizing the key results and outlining directions for future research.

2 Related works

In recent years, a variety of techniques have been proposed for brain tumor classification. These include binary classification, which differentiates between normal and tumor images, as well as multi-class classification, which identifies three types of brain tumors Glioma, Meningioma, and Pituitary, along with four-class classification, which includes normal, Glioma, Meningioma, and Pituitary.

2.1 Machine learning/deep learning for brain tumor detection: binary classification

The existing literature on brain tumor detection and classification demonstrates significant progress in both traditional machine learning (ML) and deep learning (DL) approaches. However, a common limitation is the limited critical evaluation of the strengths, weaknesses, and practical applicability of these methods. The following review provides a detailed assessment of the advantages, drawbacks, and lessons learned from recent studies. This synthesis highlights key motivations for further research in automated brain tumor analysis.

Study (Kumar et al. 2024) evaluates six classical ML algorithms—Random Forest, SVM, Logistic Regression, KNN, Naïve Bayes, and Decision Tree—combined with three feature extraction techniques: raw image loading, Histogram of Oriented Gradients (HOG), and Local Binary Patterns (LBP). Random Forest achieved the highest accuracy (99%) with raw image loading, demonstrating that classical ML can still yield strong results. A major strength of this study is the systematic comparison of classifier–feature combinations, highlighting how performance is highly sensitive to feature quality. However, handcrafted features such as HOG and LBP are insufficient to capture complex tumor morphology, resulting in inconsistent performance for KNN, Naïve Bayes, and Decision Tree. The lesson learned is that classical ML remains viable but heavily depends on feature engineering and cannot match the representational power of DL for complex medical images.

Study (Başarslan 2025) introduces the M-C&M-BL framework, which combines CNN-based spatial feature extraction with BiLSTM-based sequential modeling. With 99.33% accuracy on the Br35H dataset, the model outperformed standard CNNs such as AlexNet and BMRI-Net. The strength of this approach lies in capturing both spatial and contextual information, enhancing feature representation. Nevertheless, challenges include model generalization, computational requirements, and data privacy considerations, limiting immediate clinical deployment. The key takeaway is that hybrid DL architectures improve diagnostic accuracy but require careful validation and infrastructure support for real-world use.

Study (Panigrahi et al. 2025) evaluates multiple pretrained CNN architectures (VGG19, ResNet50V2, DenseNet201, EfficientNetB7) with additional layers and various activation functions (ReLU, PReLU, Swish). An ensemble majority-voting system achieved 99.34% accuracy. The main advantage is improved robustness and reduced misclassification through transfer learning and model fusion. However, the computational cost is high, which can limit real-time applicability in clinical settings. The lesson is that ensemble-based DL models enhance reliability but require optimization for practical deployment.

Study (Salama and Shokry 2022) proposes a framework combining a generative model for synthetic MRI creation and a classifier trained on the augmented dataset, achieving 96.88% accuracy. The key strength is addressing data scarcity and class imbalance in MRI datasets. However, synthetic data may introduce artifacts or bias, potentially

affecting clinical reliability. The lesson learned is that generative approaches are promising for data augmentation but require careful validation of synthetic image quality.

Study (Sahoo et al. 2024) presents the SSO-CCNN model using advanced hyperparameter optimization strategies combining Flower Pollination and Sunflower Optimization. The framework achieved 97.41% accuracy with fast execution (1.6 s). This approach is efficient and can improve learning through global and local search strategies. However, complex optimization algorithms may reduce transparency and reproducibility, critical factors for clinical adoption. The lesson is that optimization-based CNNs can improve performance and speed but must be balanced with interpretability and reproducibility.

Study (Özbay and Özbay 2023) employs pre-trained CNNs (DarkNet53, EfficientNet-B0, DenseNet201) with mRMR feature selection and classification using SVM, KNN, and ensemble methods. The SVM achieved 99.6% accuracy. The strength is improved discriminative power and reduced feature redundancy. However, multi-stage processing increases computational load and may pose challenges for deployment in resource-limited environments. The lesson learned is that hybrid DL-feature selection approaches improve accuracy but require computational considerations for real-world application.

Study (Hanumanthappa and Guruprakash 2024) proposes feature fusion combining handcrafted (LBP, HOG) and deep features, optimized with genetic algorithms. Canonical Correlation Analysis (CCA) achieved 93.33% accuracy. The strength is that handcrafted features can provide complementary information when combined with DL features.

However, accuracy remains lower than state-of-the-art DL models, indicating that handcrafted features alone are insufficient for complex MRI data. The lesson is that feature fusion enhances performance but deep learning must remain central.

2.2 Machine learning/deep learning for brain tumor classification: multi-classification

Brain tumor classification from MRI images has been extensively explored using both traditional machine learning (ML) and deep learning (DL) techniques. While most studies report high classification accuracy, few provide comprehensive evaluation of practical limitations, interpretability, or generalization to different datasets. Understanding the strengths and weaknesses of these approaches is essential for guiding future research and clinical adoption. Classical ML approaches using handcrafted features remain relevant.

Hybrid deep learning architectures have shown substantial improvements in tumor classification. Study (Başarslan 2025) introduced a CNN–BiLSTM framework, combining spatial feature extraction with sequential modeling, achieving 99.33% accuracy on the Br35H dataset. This hybrid approach captures both spatial and contextual information, enhancing feature representation. However, high computational requirements, limited dataset diversity, and generalization challenges remain. Similarly, Study (Panigrahi et al. 2025) evaluated multiple pre-trained CNNs (VGG19, ResNet50V2, DenseNet201, EfficientNetB7) with additional layers and ensemble majority voting, achieving 99.34% accuracy. Ensemble methods improve robustness and reduce misclassification, but their computational cost can limit real-time clinical applicability.

Other approaches have addressed data scarcity and class imbalance. Study (Salama and Shokry 2022) proposed a generative model to create synthetic MRI images combined with a classifier trained on the augmented dataset, achieving 96.88% accuracy. While this method alleviates data limitations, synthetic images may introduce artifacts or bias. Optimization-based approaches, such as the SSO-CCNN in Study (Sahoo et al. 2024), achieved 97.41% accuracy with rapid execution using metaheuristic hyperparameter tuning. These methods demonstrate efficiency and improved performance, yet their complexity may reduce transparency and reproducibility, which are critical for clinical adoption.

Combining CNN feature extraction with feature selection and fusion has also proven effective. Study (Özbay and Özbay 2023) applied mRMR feature selection to pre-trained CNN features and classified them using SVM, KNN, and ensemble methods, achieving 99.6% accuracy. Similarly, Study (Hanumanthappa and Guruprakash 2024) fused handcrafted (HOG, LBP) and deep features optimized via genetic algorithms, achieving 93.33% accuracy. These studies highlight that hybrid and fusion strategies can enhance performance, but they may increase computational complexity and remain sensitive to dataset variability.

Recent work has further refined DL and transfer learning strategies. Study (Saha et al. 2023) proposed BCM-VEMT, combining CNN-extracted features with an ensemble of ML classifiers to distinguish Glioma, Meningioma, Pituitary tumors, and Normal tissue, achieving 97.90–98.94% accuracy. Study (Remzan et al. 2024) used a feature ensemble of ResNet-50, VGG-19, and EfficientNetV2B1 with an MLP classifier, achieving 96.67% accuracy. Both studies demonstrate that ensemble and transfer learning approaches improve classification performance, but they require careful network selection and substantial computational resources.

Interpretability and clinical trust have been addressed in some studies. Study (Hosny et al. 2025) used DenseNet121 and InceptionV3 with Grad-CAM to enhance interpretability, achieving 99.02% accuracy, while Study (Ozdemir and Dogan 2024) presented MTAP, incorporating ADASYN for class imbalance, network pruning, and Avg-TopK pooling,

achieving 99.69% accuracy. Heatmap analysis showed attention on relevant brain regions, illustrating how interpretability methods provide clinical insights. However, synthetic sampling and pruning may introduce bias or reduce model transparency.

Other high-performance DL models include multi-architecture evaluations (Study Asif et al. 2023b) and dense-residual hybrids (Study Kokkalla et al. 2021), achieving accuracies of 99.67% and 99.69%, respectively. ResNet-50 with global average pooling (Study Kumar et al. 2021) reached 97.08–97.48% accuracy, demonstrating simplicity and robustness, while Study (Ghosh et al. 2024) applied EfficientNet B7 with multiple optimizers and classifiers, showing that hyperparameter selection significantly affects performance and reproducibility. Across these studies, strengths include high accuracy, robustness to noise, and effective handling of class imbalance, while limitations relate to computational cost, generalization, and interpretability.

From studies (Kumar et al. 2021; Başarslan 2025; Panigrahi et al. 2025; Salama and Shokry 2022; Sahoo et al. 2024; Özbay and Özbay 2023; Hanumanthappa and Guruprakash 2024; Saha et al. 2023; Remzan et al. 2024; Hosny et al. 2025; Ozdemir and Dogan 2024; Asif et al. 2023b; Kokkalla et al. 2021; Ghosh et al. 2024), several key lessons emerge. Deep learning consistently outperforms classical ML in modeling complex tumor morphology. Hybrid, ensemble, and feature-fusion approaches improve robustness and accuracy but increase computational requirements. Transfer learning and interpretability techniques, such as Grad-CAM, enhance clinical trust but remain underutilized. Handling class imbalance and model refinement improves accuracy, yet may introduce synthetic bias or reduce transparency. Finally, generalization across scanners, populations, and protocols is rarely validated, limiting real-world applicability.

These observations motivate the present research, which aims to develop a robust, explainable, computationally efficient, and generalizable brain tumor classification framework, bridging the gap between high research performance and practical clinical deployment.

2.3 Research gap

Despite notable progress in applying machine learning (ML) and deep learning (DL) techniques to brain tumor classification, several challenges persist. Many studies, such as those presented in Kumar et al. (2024) and (Panigrahi et al. 2025), primarily focus on binary classification (differentiating between 'normal' and 'tumor') or restricted multi-class scenarios. However, the task of accurately classifying various types of brain tumors, such as Glioma, Meningioma, and Pituitary tumors, remains insufficiently explored. While models often perform well in binary classification tasks, they struggle with multi-class classification due to issues such as class imbalance and limited dataset diversity, which negatively affect the generalizability of these models. Additionally, while pre-trained convolutional neural networks (CNNs) like VGG19, ResNet, and DenseNet (Panigrahi et al. 2025; Özbay and Özbay 2023) have been widely used for feature extraction, the integration of hybrid models and advanced feature extraction methods remains an important gap in the literature. Although generative models, such as those discussed in Salama and Shokry (2022), show potential in addressing class imbalance by generating synthetic data, challenges remain in ensuring that these synthetic datasets reflect the diversity and variability of real-world clinical data. Furthermore, the fusion of handcrafted features (e.g., LBP and HOG) with deep learning techniques, as explored in Hanumanthappa and Guruprakash (2024), holds promise but requires further optimization and integration with more sophisticated classifiers, including ensemble methods and advanced optimization algorithms like SVM and KNN. Another significant gap is the enhancement of model robustness and generalizability, particularly in clinical settings. Despite high accuracy rates reported in studies like (Panigrahi et al. 2025) and (Sahoo et al. 2024), overfitting continues to be a prevalent concern. There is also an increasing demand for model explainability, as demonstrated by the use of Grad-CAM in Panigrahi et al. (2025). However, ensuring that these models are interpretable, reliable, and applicable in clinical practice remains a critical challenge. Future research should aim to improve multi-class classification systems, address class imbalance, and refine the hybridization of models to achieve more accurate tumor detection. Moreover, integrating explainable AI and conducting clinical validation are vital steps to ensure that these models can be deployed in diverse and real-world healthcare environments.

Despite the significant progress made in multi-class brain tumor classification using deep learning and machine learning techniques, several research gaps remain. For instance, although ensemble methods, as discussed in Saha et al. (2023); Remzan et al. 2024; Hosny et al. 2025) show promising results, they generally rely on a small set of pre-trained models and classifiers. The scalability of these approaches to handle more diverse datasets, including rare or underrepresented tumor types, presents a major challenge. Furthermore, while transfer learning has been successfully employed in some studies to improve performance, there is a need for more robust and adaptive models that can seamlessly generalize across a variety of datasets without requiring extensive retraining. The integration of explainability into multi-class brain tumor classification models is another critical gap.

While techniques like Grad-CAM, used in models such as (Hosny et al. 2025), help to visualize and understand model predictions, there is still a lack of standardized frameworks for ensuring interpretability across different deep learning models. This lack of transparency hampers the adoption of these models in clinical settings, where trust in predictions is essential. Additionally, while several studies have demonstrated high accuracy in controlled environments, the performance of these models on noisy and real-world data is an ongoing concern. Models such as those in Kokkalla et al. (2021) and (Kumar et al. 2021) show promising results on noise-augmented datasets, but further research is needed to evaluate their performance under diverse imaging conditions, including variations in MRI scanners, imaging protocols, and patient demographics. Lastly, the development of models that not only achieve high accuracy but also ensure fairness and equity in multi-class classification has yet to be fully addressed. Many existing models, while accurate in general, may struggle with certain demographic groups or rare tumor types, leading to potential misclassifications in clinical practice. Future research should focus on enhancing the fairness, generalizability, and robustness of multi-class brain tumor classification systems, closing these gaps to enable the successful integration of AI-powered diagnostic tools into clinical workflows.

3 Dataset and proposed method

In this section, we describe the dataset utilized in our study and detail the proposed methodology for brain tumor classification.

3.1 Dataset

The dataset used in this study, referred to as the "brain-tumor-mri-dataset," was obtained from the Kaggle open data repository (https://www.kaggle.com, datasets, Masoud Nickparvar, brain-tumor-mri-dataset). It contains a total of 7023 MRI images, divided into two main folders: Training (5712 images) and Testing (1311 images). The training set is further categorized into four classes: glioma (1321 images), meningioma (1339 images), no tumor (1595 images), and pituitary (1457 images). The testing set follows a similar structure, with glioma (300 images), meningioma (306 images), no tumor (405 images), and pituitary (300 images). This balanced class distribution supports reliable training and evaluation of classification models by ensuring adequate representation of each category. Figure 1 shows sample images from the dataset, illustrating the visual characteristics of each tumor type.

3.2 Proposed method

The proposed method for brain tumor classification in MRI images, as illustrated in Fig. 2, begins with image preprocessing and data augmentation to improve model robustness and generalization. MRI scans are resized and augmented using geometric transformations, including horizontal and vertical flipping, which allow the model to learn orientation-invariant features, and photometric adjustments such as random changes in brightness, contrast, saturation, and hue, which simulate real-world variability in MRI acquisition. These augmentations ensure that the model focuses on discriminative tumor characteristics rather than imaging artifacts or specific acquisition conditions. The augmented images are subsequently processed through several pre-trained convolutional neural networks (CNNs) for feature extraction, namely VGG16, VGG19, ResNet50, ResNet152, and EfficientNetB3. Each model is chosen for its complementary strengths: VGG16 and VGG19 extract hierarchical spatial features with a simple and effective architecture; ResNet50 and ResNet152 utilize residual connections to enable very deep networks to learn complex and abstract representations; EfficientNetB3 incorporates Mobile Inverted Bottleneck (MBConv) blocks with Squeeze-and-Excitation modules, providing multi-scale feature extraction with computational efficiency. The extracted features are reshaped and fed into a Long Short-Term Memory (LSTM) network, which captures sequential dependencies across the feature maps. The LSTM employs sigmoid and tanh activation functions to model non-linear dependencies and maintain long-range feature correlations. Batch normalization, dropout, and dense layers are applied after the LSTM to refine the features, prevent overfitting, and enhance generalization. For the final classification step,

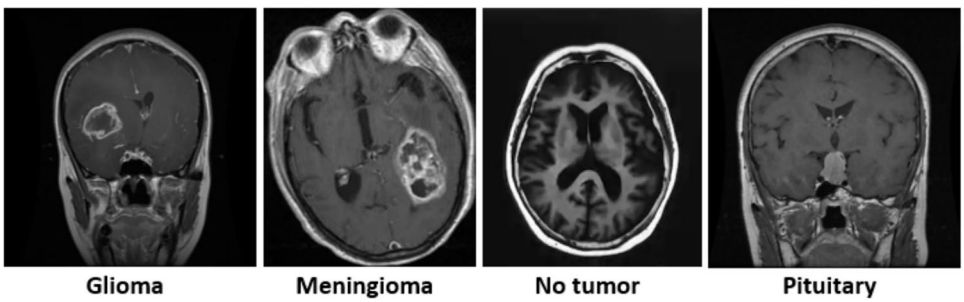


Fig. 1 Sample images from the dataset

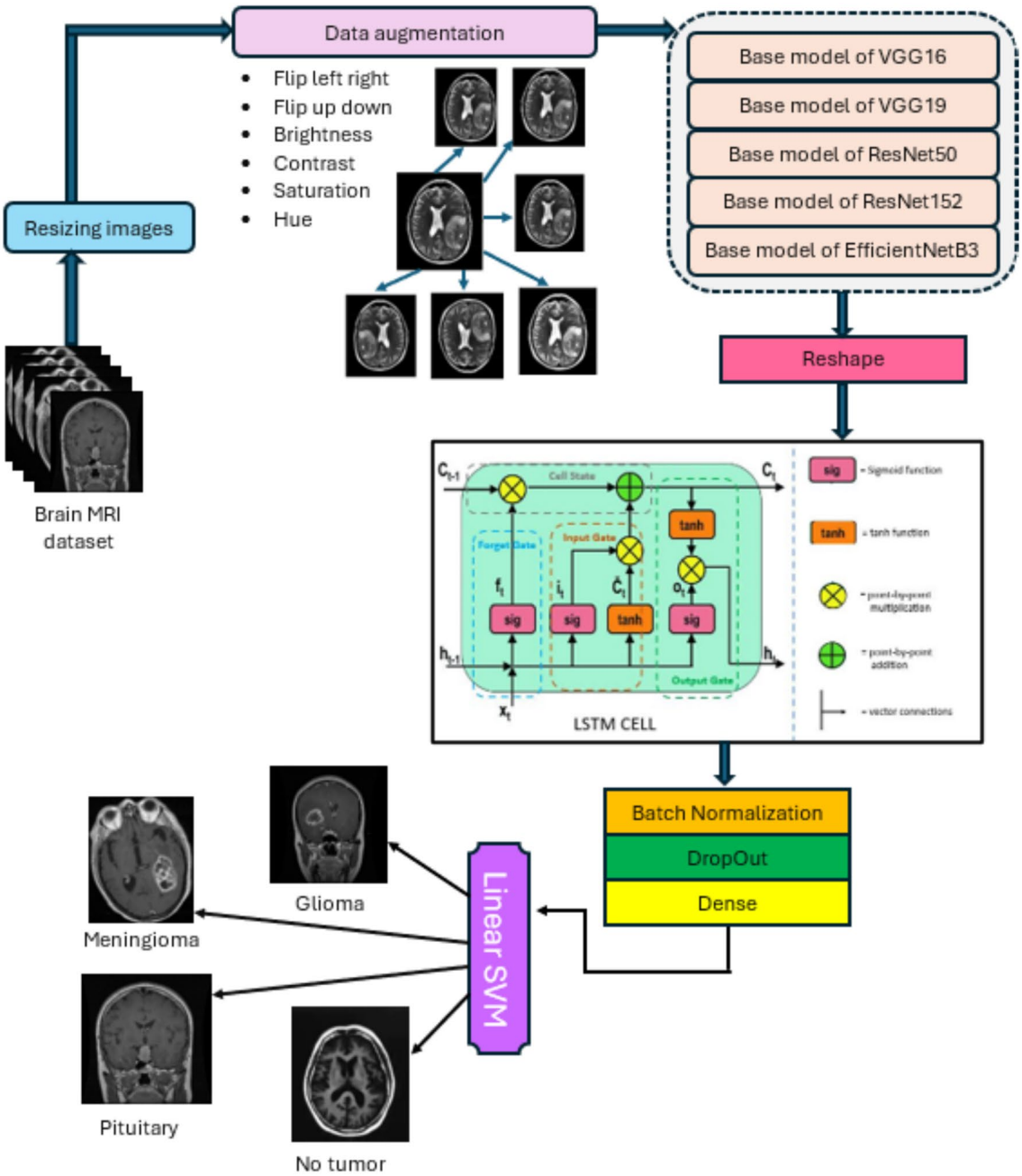


Fig. 2 Proposed method

a Linear Support Vector Machine (SVM) is employed. The SVM is selected for its ability to construct optimal hyperplanes in high-dimensional feature spaces, providing robust margin-based separation of the four classes: Meningioma, Glioma, Pituitary, and No Tumor. By combining deep CNN-based feature extraction, sequential modeling with LSTM, and SVM classification, the proposed hybrid framework effectively addresses challenges such as complex feature representation, sequential dependencies, overfitting, and class separation, resulting in a robust and accurate brain tumor classification system.

- Data augmentation

Data augmentation was applied during the preprocessing stage using TensorFlow's image transformation functions. The augmentation pipeline integrates both geometric and photometric transformations to enhance model robustness and simulate real-world variability. Geometric augmentations, including random horizontal and vertical flipping, allow the model to learn orientation-invariant features. Photometric augmentations further simulate acquisition variability: random brightness adjustments (maximum delta = 0.1)

account for illumination changes; contrast and saturation are randomly varied within the range of 0.8–1.2 to reflect differences in contrast and color intensity; and random hue shifts (maximum delta = 0.1) simulate variations in color tones. The complete augmentation workflow is summarized in Algorithm 1, showing how input images are systematically transformed to improve generalization by encouraging the model to focus on discriminative features rather than specific imaging conditions or tumor orientations.

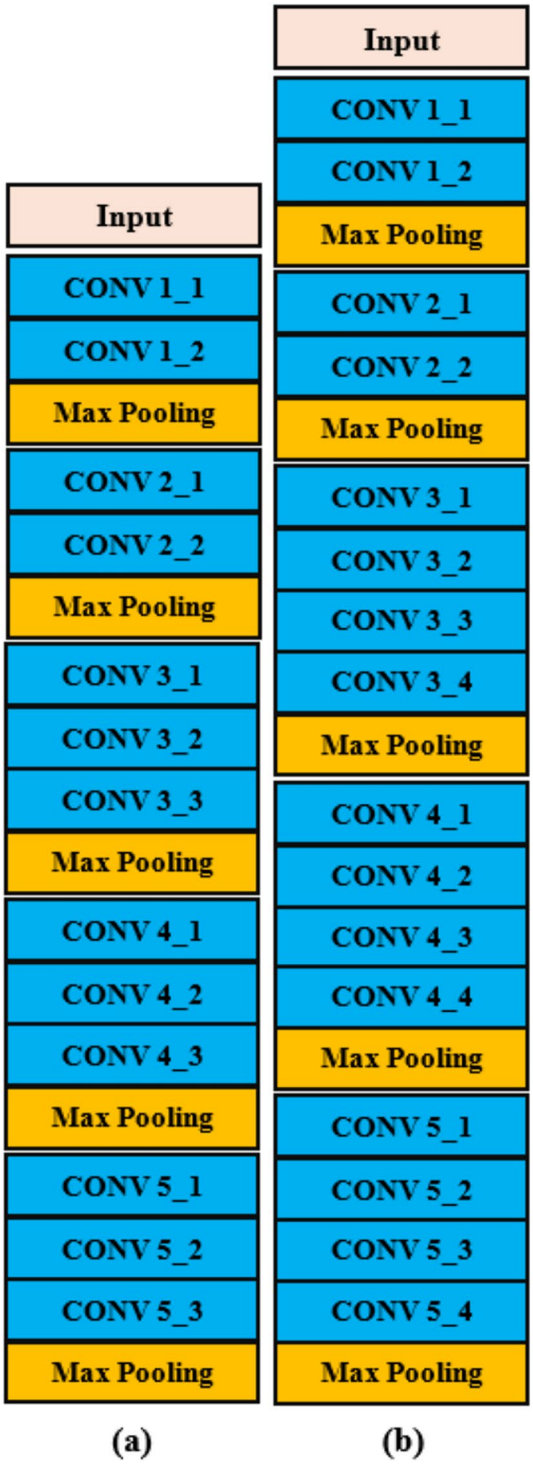


Fig. 3 Base models used for feature extraction: (**a**) VGG16 and (**b**) VGG19

Algorithm 1 Data Augmentation Workflow for Tumor Image Preprocessing

1. **Input Image**
   Original dataset image (tumor scan).
2. **Geometric Augmentations**
   - Random horizontal flip
   - Random vertical flip
3. **Photometric Augmentations**
   - Random brightness adjustment (max delta = 0.1)
   - Random contrast adjustment (range 0.8–1.2)
   - Random saturation adjustment (range 0.8–1.2)
   - Random hue shift (max delta = 0.1)
4. **Output Augmented Image**
   Image with variations in orientation, illumination, contrast, saturation, and hue, enhancing generalization.

- Base Models for Feature Extraction: VGG16 and VGG19

The base models used for feature extraction are VGG16 and VGG19, which are deep convolutional neural networks recognized for their simplicity and effectiveness. Both models consist of a series of convolutional layers with small receptive fields (3×3 filters), followed by max-pooling layers. The key difference between them is the number of convolutional layers: VGG16 has 16 weight layers, while VGG19 has 19. Both architectures are commonly pre-trained on large image datasets such as ImageNet, and they are widely used for transfer learning due to their capability to extract rich feature representations from images. These models serve as the foundation for the subsequent classification process in our method. Figure 3 illustrates the architecture of VGG16 and VGG19.

- Base Models for Feature Extraction: ResNet50 and ResNet152

In this study, ResNet50 and ResNet152 are employed as base models for feature extraction. Both are based on the ResNet architecture, which leverages residual connections to mitigate the vanishing gradient problem and enable the training of very deep neural networks. ResNet50, with 50 layers, offers a good trade-off between accuracy and computational efficiency, making it suitable for resource-constrained applications. In contrast, ResNet152, with 152 layers, provides greater representational power by capturing more complex and abstract features, albeit at a higher computational cost. These models are well suited for transfer learning and can

be fine-tuned for tasks such as image classification, object detection, and other computer vision applications.

Table 1presents a comparison of their architectures. While both networks share the same initial convolutional layer and pooling strategy, they differ in the number of residual blocks. ResNet50 employs fewer block repetitions, whereas ResNet152 introduces a much deeper configuration, particularly in the conv4_x stage, where it includes 36 residual blocks compared to only 6 in ResNet50. This increased depth allows ResNet152 to extract more detailed feature representations, as reflected in its higher computational complexity (11.3×10^9 FLOPs vs. 3.8×10^9 FLOPs for ResNet50).

- Base Model for Feature Extraction: EfficientNetB3

The EfficientNetB3 architecture is employed as the backbone for feature extraction. As illustrated in Fig. 4, the model comprises a series of computationally efficient Mobile Inverted Bottleneck (MBConv) blocks. The macro-architecture, shown in Fig. 4(a), begins with a standard 3×3 convolution, followed by multiple sequential stages of MBConv blocks with varying expansion ratios, kernel sizes, and strides to progressively extract and refine multi-scale features. The network concludes with a 1×1 convolution, global pooling, and a final fully-connected layer for classification. The detailed structure of the fundamental MBConv block with an expansion factor of 6 and a 5×5 kernel is depicted in Fig. 4(b). Its operation involves an initial 1×1 convolution to expand the channel dimension, followed by a 5×5 depthwise convolution, Batch Normalization (BN), and a Swish activation. A Squeeze-and-Excitation (SEM) module is then applied to model channel-wise dependencies, after which a 1×1 convolution projects the features back to a lower dimension. A residual connection skips this entire block when the input and output dimensions match.

- LSTM (Long Short-Term Memory) Network

LSTM (Long Short-Term Memory) networks are a type of recurrent neural network (RNN) designed to address the vanishing gradient problem, enabling the model to capture long-term dependencies in sequential data. LSTMs are particularly effective for tasks such as time-series forecasting, speech recognition, and natural language processing. Unlike traditional RNNs, which struggle to learn long-term dependencies, LSTMs are equipped with memory cells and gating mechanisms that control the flow of information. These gates—specifically the forget gate, input gate, and output gate—regulate the information retained in the memory cell at each time step.

At each time step, the forget gate determines what information from the previous cell state should be discarded:

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \tag{1}$$

where σ is the sigmoid activation function, W_f is the weight matrix for the forget gate, h_{t-1} is the previous hidden state, and x_t is the current input.

The input gate controls which new information should be added to the cell state

$$i_t = \sigma\left(W_i \cdot [h_{t-1}, x_t] + b_i\right) \tag{2}$$

Table 1 Comparative architecture of ResNet50 and ResNet152

Stage	ResNet50	ResNet152
conv1 (112×112)	7×7, 64, stride 2	
conv2_x (56×56)	3×3 max pool, stride 2	
	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix} \times 3$
conv3_x (28×28)	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1\times1, 128; \\ 3\times3, 128; \\ 1\times1, 512 \end{bmatrix} \times 8$
conv4_x (14×14)	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1\times1, 256; \\ 3\times3, 256; \\ 1\times1, 1024 \end{bmatrix} \times 36$
conv5_x (7×7)	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times1, 64; \\ 3\times3, 64; \\ 1\times1, 2048 \end{bmatrix} \times 3$
Final Layer	average pool, 1000-d fc, softmax	
FLOPs	3.8×10^9	11.3×10^9

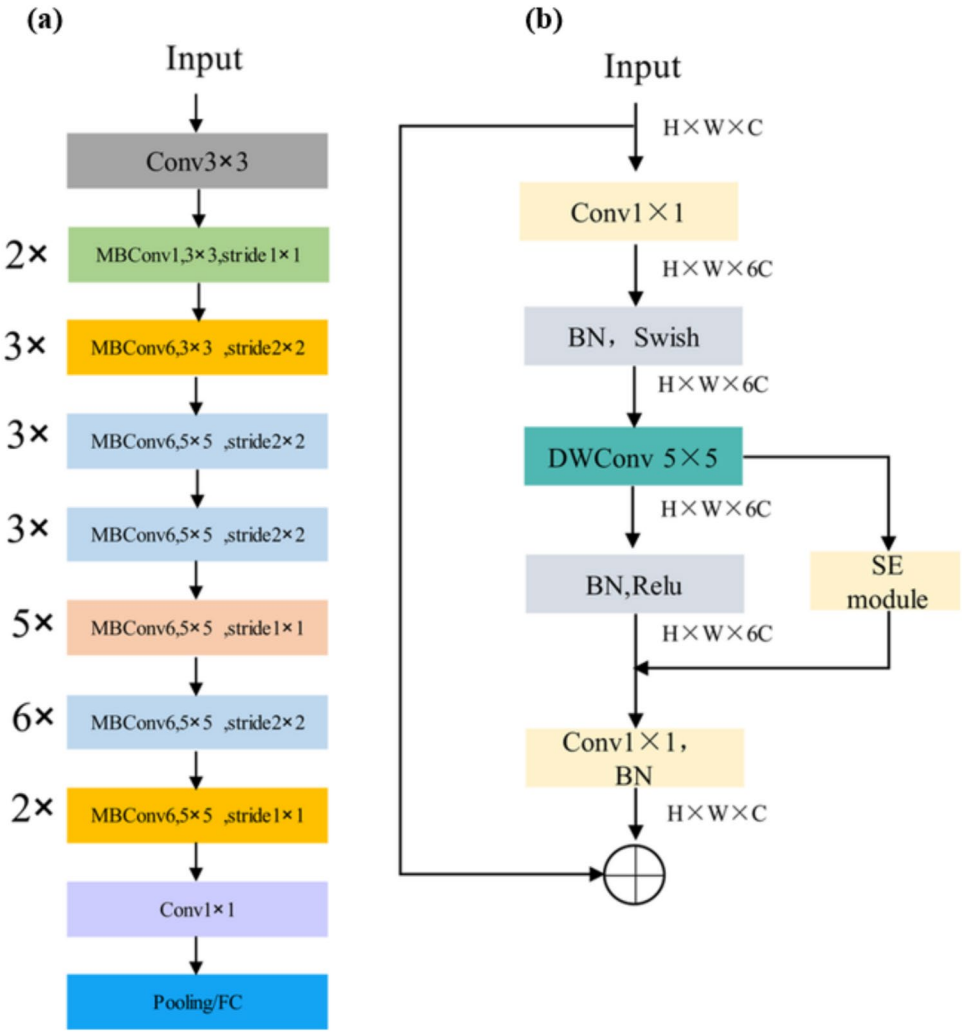


Fig. 4 EfficientNetB3 structure: (**a**) EfficientNetB3, (**b**) MBConv6 5 × 5

where W_i is the weight matrix for the input gate, and b_i is the bias term.

The candidate cell state $\widetilde{C}_t$ is created using the hyperbolic tangent function tanh, and this will be added to the cell state

$$\widetilde{C}_t = tanh\left(W_C \cdot [h_{t-1}, x_t] + b_C\right) \tag{3}$$

where W_C is the weight matrix for the candidate state and b_C is the bias.

The cell state C_t is updated by combining the previous cell state, scaled by the forget gate, and the candidate cell state, weighted by the input gate

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \widetilde{C}_t \tag{4}$$

where C_{t-1} is the previous cell state

The output gate regulates which part of the cell state should be passed to the hidden state:

$$o_t = \sigma\left(W_o \cdot [h_{t-1}, x_t] + b_0\right) \tag{5}$$

where W_o is the weight matrix for the output gate, and b_0 is the bias term.

Finally, the hidden state h_t is the output of the LSTM, which is produced by applying the output gate to the cell state

$$h_t = o_t \cdot \tanh(C_t) \tag{6}$$

The hidden state h_t is used as the output of the LSTM and is passed to the next time step.

By using these gating mechanisms, LSTMs can effectively learn to capture long-range dependencies, making them well-suited for tasks requiring the understanding of temporal or sequential patterns.

- Training a Linear SVM Classifier on PCA-Reduced and Standardized Features

To train the Support Vector Machine (SVM) classifier on the extracted features, a preprocessing pipeline was applied. First, feature standardization was performed using the StandardScaler to normalize the data with zero mean and unit variance, which is essential for the performance of SVMs. Principal Component Analysis (PCA) was then used to reduce the dimensionality of the feature space, helping to minimize redundancy and computational cost. The number of components was set to the minimum of the number of samples, number of features, and 64, ensuring an optimal balance between information retention and overfitting prevention. The standardized features were transformed using PCA, and the resulting components were used to train a linear SVM with a regularization parameter $C=1.0$. Probability estimation was enabled to allow for probabilistic output.

The linear SVM seeks to find the optimal hyperplane defined by:

$$f(x) = w^T x + b \tag{7}$$

where w is the weight vector and b is the bias. The training process involves solving the following convex optimization problem:

$$min_{w,b} \frac{1}{2}\|w\|^2 \; subject\ to\ y_i\left(w^T x_i + b\right) \geq 1, \forall i \tag{8}$$

This approach ensures that the SVM is trained on a compact, normalized representation of the data, improving both classification performance and generalization.

4 Results

4.1 Performance evaluation

To measure the performance of this classification, it is necessary to establish a distinction between two types of classified elements within the desired category: TP (True Positive), FP (False Positive), FN (False Negative), TN (True Negative).

4.1.1 Accuracy

The accuracy refers to the ratio of correctly predicted instances to the total instances in a dataset. It's a measure used to evaluate the performance of a model. Mathematically, accuracy is calculated as:

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \tag{9}$$

4.1.2 Recall

Recall, also known as sensitivity or true positive rate, is a performance metric in classification that measures the ability of a model to correctly identify all relevant instances from a dataset. It's particularly important in scenarios where the identification of a specific class or condition is crucial. Mathematically, recall is calculated as:

$$Recall = \frac{TP}{FN + TP} \tag{10}$$

4.1.3 Precision

Precision is a performance metric in classification that measures the accuracy of the positive predictions made by a model. It focuses on the relevance of the predicted positive cases, indicating how many of the instances classified as positive are indeed relevant. Mathematically, precision is calculated as:

$$Precision = \frac{TP}{FP + TP} \tag{11}$$

4.1.4 F1 score

Comparing two models with low precision and high recall (or vice versa) can be challenging. Therefore, to make them comparable, we use the F1 Score. It measures both recall and precision simultaneously. It employs the harmonic mean instead of the arithmetic mean, penalizing extreme values more effectively.

$$\text{F1 score} = 2 \times \frac{precision \times recall}{precision + recall} \tag{12}$$

4.2 Hyperparameters used in deep learning models

Proper selection of hyperparameters is essential for achieving optimal performance and stable convergence in deep learning models. Table 2 summarizes the key hyperparameters applied to the proposed CNN + LSTM + SVM architectures. Four widely used optimizers: Adam, SGD, AdaGrad, and RMSprop, were employed to evaluate their influence on training stability and convergence speed. A learning rate of 1e-4 was chosen to provide a balance between convergence efficiency and gradient stability, reducing the risk of overshooting minima.

The categorical cross-entropy loss function was adopted to optimize the multi-class classification task, as it effectively quantifies the discrepancy between predicted probabilities and true labels. A maximum of 30 epochs ensured

Table 2 Hyperparameter configuration for deep learning models

Parameter	Value
Optimizer algorithm	Adam, SGD, AdaGrad, RMSprop
Learning Rate	1e-4
Loss function	Categorical cross entropy
Maximum epoch	30
Batch size	32
Drop out	0.3

adequate training while limiting overfitting. A batch size of 32 offered a good trade-off between computational efficiency and gradient estimation accuracy. Additionally, a dropout rate of 0.3 was applied to the fully connected layers to enhance generalization and prevent overfitting. This hyperparameter configuration collectively supports stable training, effective feature learning, and robust performance in classifying brain tumors.

4.3 Results

This section presents a comparative evaluation of five hybrid deep learning architectures (VGG16+LSTM+SVM, VGG19+LSTM+SVM, ResNet50+LSTM+SVM, ResNet152+LSTM+SVM, and EfficientNetB3+LSTM+SVM) trained with different optimizers, namely Adam, SGD, AdaGrad, and RMSprop. To assess their performance, confusion matrices were generated for each case, comparing the predicted labels against the true labels for the four tumor categories: glioma, meningioma, no tumor, and pituitary tumor. These matrices allow a detailed inspection of correct classifications (diagonal values) and misclassifications (off-diagonal values), thereby providing insight into each model's discriminative power and robustness across categories.

Figure 5 presents the confusion matrices for five deep learning architectures combined with LSTM and SVM classifiers for multi-class brain tumor classification. The evaluated models include VGG16+LSTM+SVM, VGG19+LSTM+SVM, ResNet50+LSTM+SVM, ResNet152+LSTM+SVM, and EfficientNetB3+LSTM+SVM, all trained using the Adam optimizer. The matrices compare the true labels (rows) with the predicted labels (columns) across four tumor categories: glioma, meningioma, no tumor, and pituitary tumor. Overall, the results indicate strong classification performance for all models, as evidenced by the high values along the diagonal of each matrix, representing correct predictions. The off-diagonal values highlight misclassifications, revealing which tumor types are more frequently confused. The VGG16+LSTM+SVM and VGG19+LSTM+SVM models exhibit comparable performance; however, VGG19 shows slightly higher misclassification rates, particularly between glioma and meningioma. This suggests that, despite its increased depth, VGG19 may still struggle to capture the subtle distinctions between visually similar tumor types. The ResNet-based models provide noticeable improvements, with ResNet152 achieving fewer misclassifications than ResNet50. This enhancement can be attributed to the greater depth and residual connections of ResNet152, which enable more effective extraction of complex imaging features. Among all evaluated architectures, EfficientNetB3+LSTM+SVM delivers the highest accuracy and the lowest number of misclassifications. Its superior performance is likely due to its compound scaling strategy, which balances depth, width, and resolution to produce highly expressive yet computationally efficient feature representations. In summary, while all models demonstrate strong classification capability, the results highlight the advantages of advanced architectures—particularly ResNet152 and EfficientNetB3—in improving accuracy and reducing misclassification in brain tumor classification tasks. These findings emphasize the importance of selecting an optimized model architecture for achieving robust performance in medical imaging applications.

Figure 6 presents the confusion matrices for five deep learning models combined with LSTM and SVM classifiers for multi-class tumor classification using the Stochastic Gradient Descent (SGD) optimizer. Overall, the results indicate that all models achieve reasonable performance, as shown by the concentration of values along the diagonal. However, notable differences in accuracy and error distribution are observed across the architectures. The VGG16+LSTM+SVM model (Fig. 6(a)) shows moderate performance, with substantial misclassifications between glioma and meningioma, where 72 glioma cases are incorrectly predicted as meningioma and 40 meningioma cases are misclassified as glioma. This highlights the model's limited ability to capture subtle structural differences between these two tumor types. The VGG19+LSTM+SVM model (Fig. 6(b)) offers only marginal improvement, as misclassifications between glioma and meningioma remain frequent, indicating that increased depth alone is insufficient when using SGD. The ResNet-based models demonstrate clearer improvements. ResNet50+LSTM+SVM (Fig. 6(c)) produces fewer misclassifications, with 238 correct predictions for glioma and 224 for meningioma, although some confusion persists between meningioma and the no-tumor category. ResNet152+LSTM+SVM (Fig. 6(d)) shows further improvement, with higher correct predictions for glioma (238), meningioma (224), and pituitary tumor (281), suggesting that deeper residual architectures enhance feature extraction and generalization. Among all evaluated models, EfficientNetB3+LSTM+SVM (Fig. 6(e)) achieves the highest performance. It correctly predicts 386 no-tumor

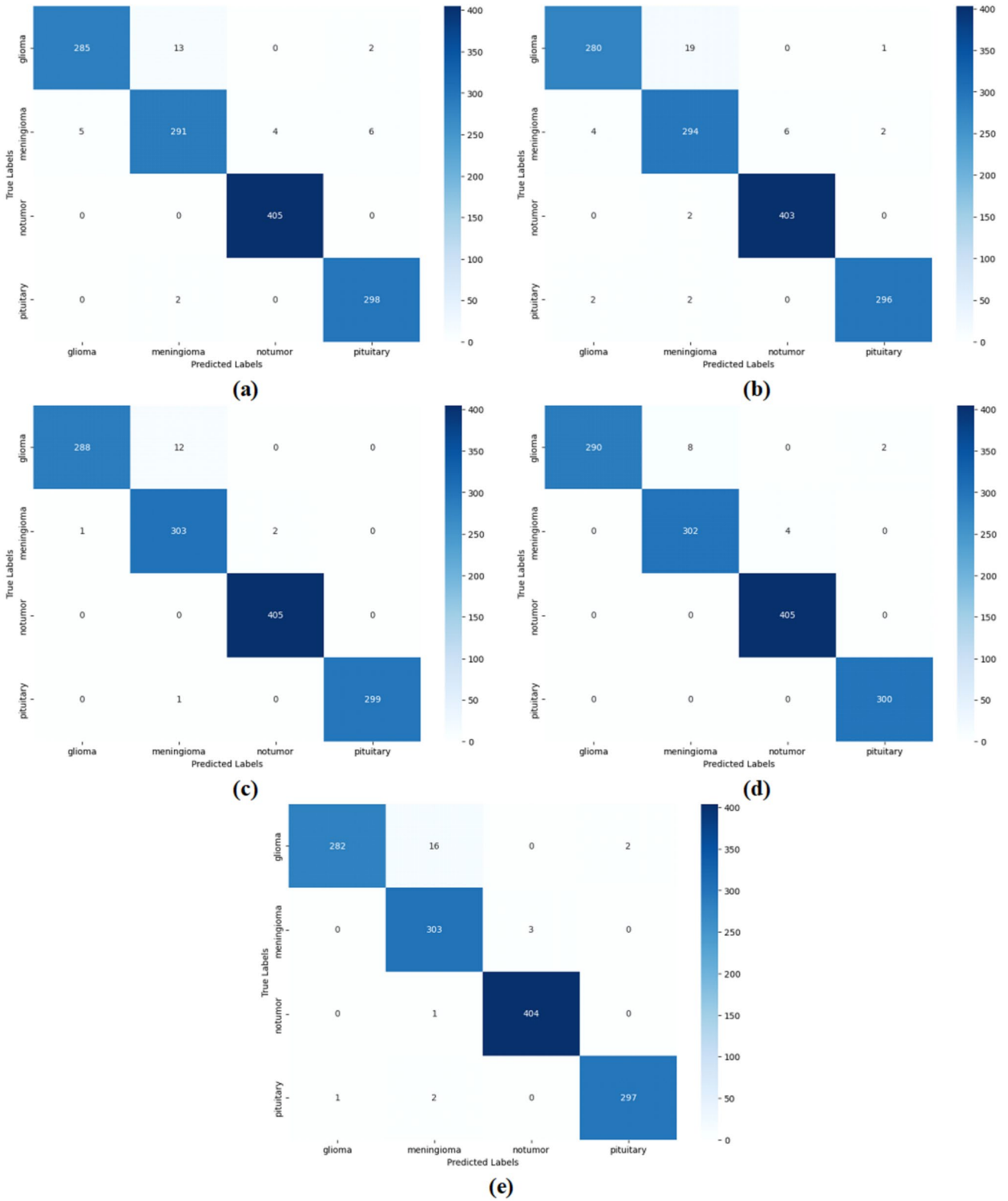


Fig. 5 Confusion matrices for (**a**) VGG16 + LSTM + SVM, (**b**) VGG19 + LSTM + SVM, (**c**) ResNet50 + LSTM + SVM, (**d**) ResNet152 + LSTM + SVM, and (**e**) EfficientNetB3 + LSTM + SVM using adam optimizer

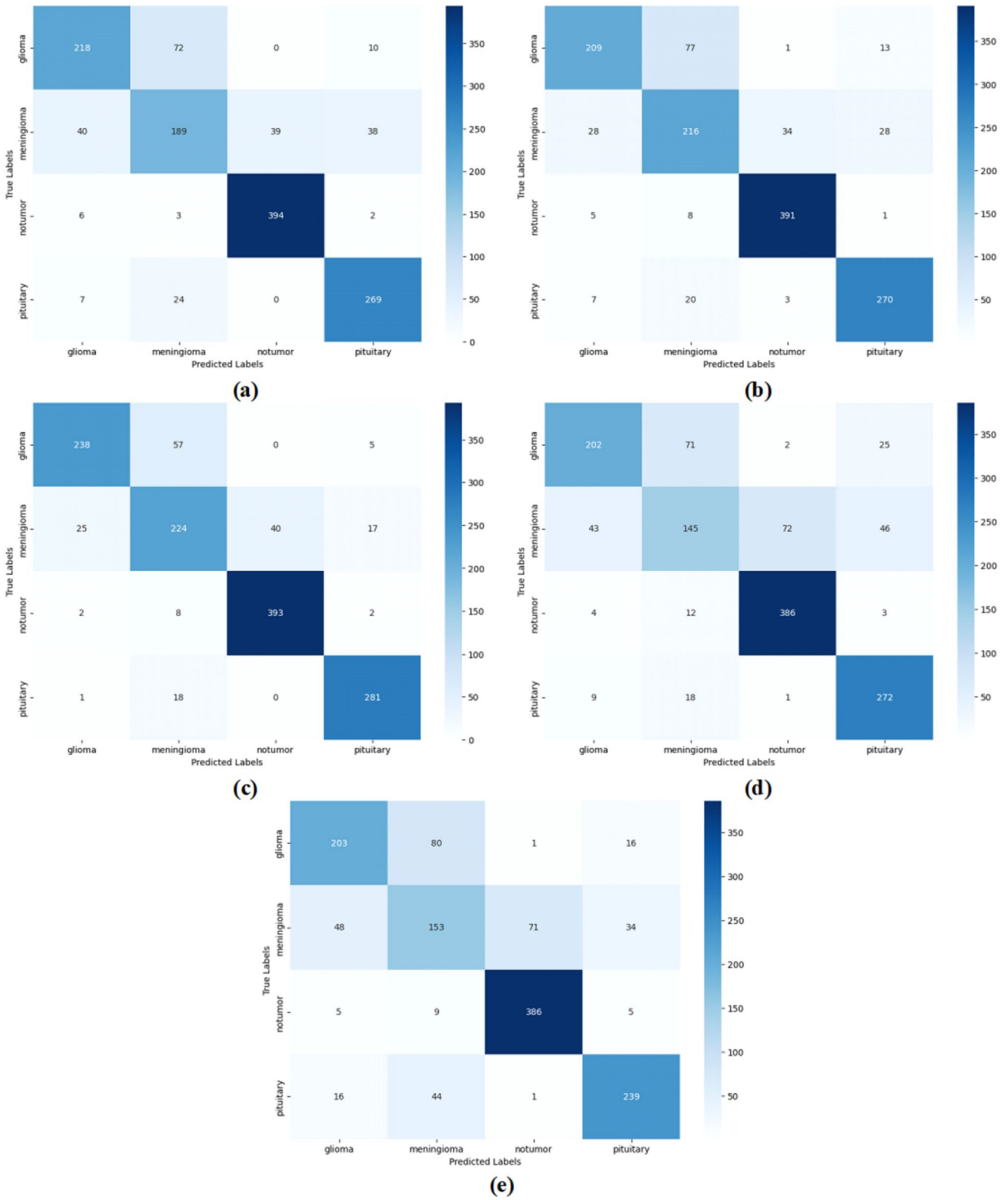


Fig. 6 Confusion matrices for (**a**) VGG16 + LSTM + SVM, (**b**) VGG19 + LSTM + SVM, (**c**) ResNet50 + LSTM + SVM, (**d**) ResNet152 + LSTM + SVM, and (**e**) EfficientNetB3 + LSTM + SVM using SGD optimizer

cases and records the lowest misclassification rates overall, despite some remaining confusion between glioma and meningioma. This performance can be attributed to EfficientNet's compound scaling strategy, which optimally balances depth, width, and resolution, enabling more effective feature learning. In conclusion, while all models demonstrate acceptable classification capability with the SGD optimizer, deeper and more efficient architectures—particularly EfficientNetB3—provide superior accuracy and reduced misclassification rates. These results highlight the importance of selecting advanced model architectures for complex medical image classification tasks, where distinguishing subtle differences between tumor types is essential for achieving reliable diagnostic outcomes.

Figure 7 presents the confusion matrices for five deep learning models combined with LSTM and SVM classifiers for multi-class tumor classification using the AdaGrad optimizer. The VGG16+LSTM+SVM model (Fig. 7(a)) demonstrates reasonable performance, achieving 223 correct predictions for glioma and 279 for pituitary tumor. However, it shows substantial misclassifications between glioma and meningioma, with 60 glioma cases incorrectly predicted as meningioma and 39 meningioma cases misclassified as glioma. Although the model performs well on the no-tumor class (391 correct predictions), these results indicate that VGG16 struggles to capture the subtle visual similarities between glioma and meningioma, suggesting that its feature extraction capacity may be insufficient for this challenging distinction. The VGG19+LSTM+SVM model (Fig. 7(b)) offers slight improvement over VGG16, achieving 216 correct glioma predictions and 395 correct pituitary tumor predictions. Nevertheless, the misclassification trend between glioma and meningioma persists, with 69 glioma cases misclassified as meningioma and 38 meningioma cases misclassified as glioma. While deeper than VGG16, VGG19 does not achieve a major performance gain, likely due to the limitations of AdaGrad, which rapidly decreases the learning rate and may prevent deeper networks from fully optimizing their representations. The ResNet50+LSTM+SVM model (Fig. 7(c)) shows a more substantial improvement. With 249 correct predictions for glioma and 236 for meningioma, the model significantly reduces misclassifications—only 31 meningiomas are misclassified as glioma and 19 gliomas as meningioma. Performance also improves for the no-tumor class (392 correct predictions) and the pituitary tumor class (281 correct predictions). These results demonstrate the advantage of residual connections, which allow deeper feature extraction and enable the model to capture more complex structural and textural tumor characteristics. The ResNet152+LSTM+SVM model (Fig. 7(d)) further enhances performance, with 249 correct predictions for glioma, 236 for meningioma, and 392 for the no-tumor class. Misclassifications between glioma and meningioma are further reduced, with 47 glioma cases misclassified as meningioma and 20 meningioma cases misclassified as glioma. This model provides the highest accuracy among the architectures evaluated so far, confirming that deeper residual networks offer improved generalization and better separation between visually similar tumor categories. Finally, the EfficientNetB3+LSTM+SVM model (Fig. 7(e)) outperforms all other models, achieving strong results across all tumor types. It achieves 249 correct predictions for glioma, 236 for meningioma, and 392 for the no-tumor class, while showing minimal misclassifications—47 glioma cases misclassified as meningioma and 31 meningioma cases misclassified as glioma. EfficientNetB3's superior performance can be attributed to its compound scaling strategy, which optimally balances network depth, width, and resolution, enabling more efficient and expressive feature representation. This results in the highest classification accuracy and the fewest misclassifications among all models analyzed. In conclusion, while all models achieve good classification performance, the results reveal a clear pattern: deeper and more advanced architectures such as ResNet152 and, especially, EfficientNetB3 consistently outperform shallower VGG models. These findings highlight the importance of employing efficient and capable feature extractors for complex medical imaging tasks, where subtle distinctions between tumor types must be accurately captured to support reliable diagnostic decisions. Future improvements may include exploring alternative optimizers, integrating attention mechanisms, or expanding the dataset to further enhance classification robustness.

Figure 8 illustrates the confusion matrices obtained from the classification results of five hybrid architectures—VGG16+LSTM+SVM, VGG19+LSTM+SVM, ResNet50+LSTM+SVM, ResNet152+LSTM+SVM, and EfficientNetB3+LSTM+SVM—optimized using the RMSprop optimizer. The VGG16+LSTM+SVM model (Fig. 8(a)) shows some misclassification between glioma and meningioma, with 26 glioma cases predicted as meningioma, and 27 meningioma cases misclassified as no tumor. In contrast, the no-tumor and pituitary tumor classes are predicted with high accuracy (403/405 and 293/300, respectively), indicating that the model is more effective for classes with distinct visual features. The VGG19+LSTM+SVM model (Fig. 8(b)) slightly improves meningioma classification, although misclassifications remain: 28 gliomas are misclassified as meningiomas, and some meningiomas are predicted as pituitary tumors. The no-tumor and pituitary classes continue to exhibit high accuracy, suggesting that deeper VGG architectures provide marginal benefits for distinguishing visually similar tumors under RMSprop optimization. The ResNet50+LSTM+SVM model (Fig. 8(c))

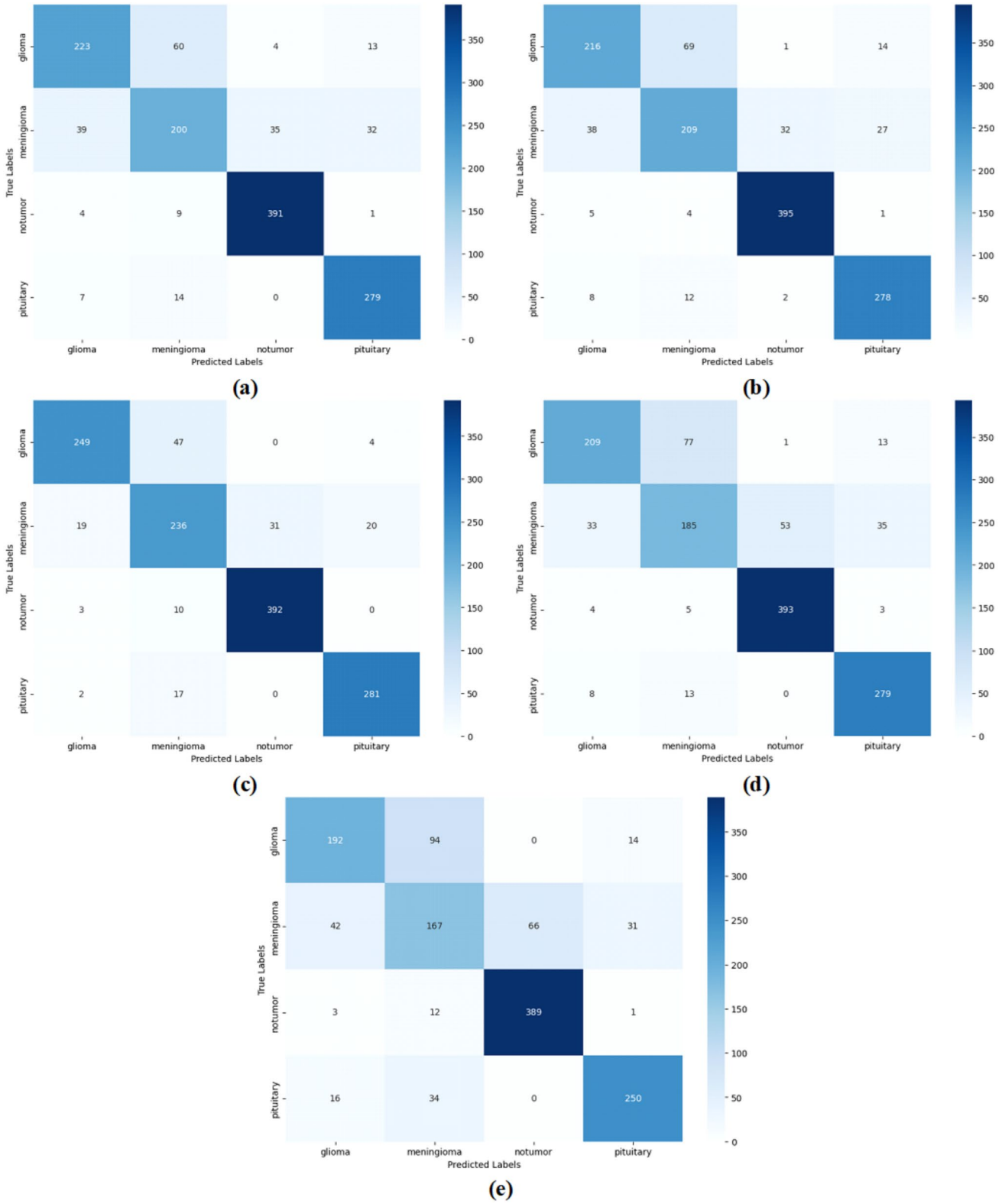


Fig. 7 Confusion matrices for (**a**) VGG16+LSTM+SVM, (**b**) VGG19+LSTM+SVM, (**c**) ResNet50 + LSTM + SVM, (**d**) ResNet152+LSTM+SVM, and (**e**) EfficientNetB3+LSTM+SVM using AdaGrad optimizer

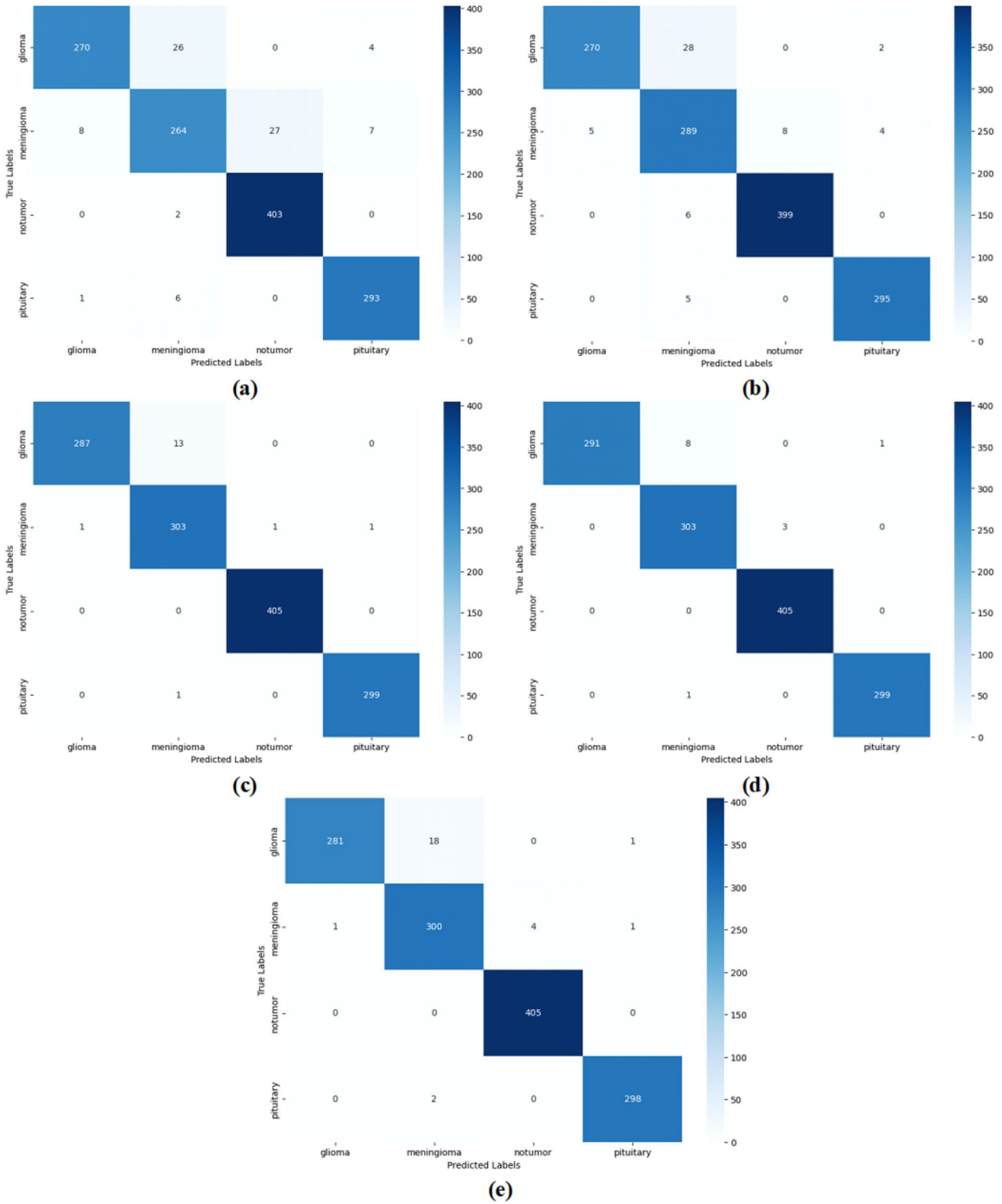


Fig. 8 Confusion matrices for (**a**) VGG16+LSTM+SVM, (**b**) VGG19+LSTM+SVM, (**c**) ResNet50 + LSTM + SVM, (**d**) ResNet152+LSTM+SVM, and (**e**) EfficientNetB3+LSTM+SVM Using RMSprop optimizer

demonstrates improved separation across classes, with only 13 gliomas misclassified as meningiomas and a single meningioma misclassified as pituitary. Notably, the no-tumor class achieves perfect recognition (405/405), highlighting the strength of residual networks in capturing discriminative features and distinguishing between normal and pathological cases. The ResNet152+LSTM+SVM model (Fig. 8(d)) achieves the best overall performance, with minimal misclassifications: 8 gliomas misclassified as meningiomas and 3 meningiomas as no tumor, while all other instances are correctly classified. This demonstrates the advantages of deeper residual networks in enhancing feature extraction and improving differentiation between tumor types with overlapping imaging characteristics. The EfficientNetB3+LSTM+SVM model (Fig. 8(e)) also produces competitive results. Misclassifications are slightly higher (18 gliomas misclassified as meningiomas and 4 meningiomas as no tumor), but both the no-tumor and pituitary classes are nearly perfectly recognized. This reflects EfficientNetB3's ability to balance high accuracy with computational efficiency, leveraging its optimized architecture to capture complex patterns.

Overall, all models demonstrate strong classification capability, as shown by the concentration of values along the diagonal in the confusion matrices. Among them, ResNet152+LSTM+SVM achieves the most reliable results with the lowest misclassification rates, followed by ResNet50 and EfficientNetB3. VGG-based models, while effective, exhibit more confusion between glioma and meningioma, indicating limitations in capturing subtle inter-class differences. These results underscore the importance of deeper and more advanced architectures for accurate brain tumor classification. Across all optimizers (Adam, SGD, AdaGrad, and RMSprop), the confusion matrix analyses confirm that the proposed hybrid architectures achieve robust performance, with most samples correctly identified. The most frequent errors occur between glioma and meningioma, reflecting the clinical challenge of distinguishing these tumor types due to overlapping imaging characteristics. ResNet152+LSTM+SVM and EfficientNetB3+LSTM+SVM consistently deliver superior results, exhibiting minimal misclassifications and higher overall reliability. These findings demonstrate that deeper and more efficient architectures enhance discriminative power, reinforcing their potential for accurate, automated brain tumor diagnosis.

The comprehensive results presented in Table 3 provide critical insights into the performance of hybrid CNN + LSTM + SVM models, highlighting a decisive interdependence between the convolutional backbone architecture and the choice of optimization algorithm. The data reveals a clear performance hierarchy among the optimizers, with Adam and RMSprop consistently enabling superior model efficacy, while SGD and AdaGrad resulted in significantly diminished performance across all evaluated metrics.

A detailed examination shows that the Adam optimizer facilitated the highest overall performance, with model accuracies ranging from 97 to 99%. The ResNet-based architectures, specifically ResNet50 and ResNet152, achieved peak performance with 99% accuracy, accompanied by precision, recall, and F1 scores all exceeding 98.75%. This indicates not only high predictive power but also a well-balanced performance across classes. The EfficientNetB3 and VGG models also demonstrated strong results under Adam, with

Table 3 Comparative performance metrics of hybrid CNN + LSTM + SVM models across different optimizers

Model	Accuracy	Precision	Recall	F1 score	Optimizer
VGG16+LSTM+SVM	98%	97.25%	97.25%	97.5%	Adam
VGG19+LSTM+SVM	97%	97.25%	97%	97%	
ResNet50+LSTM+SVM	99%	99%	98.75%	98.75%	
ResNet152+LSTM+SVM	99%	98.75%	99%	99%	
EfficientNetB3+LSTM+SVM	98%	98%	98%	98%	
VGG16+LSTM+SVM	82%	80.25%	80.5%	80.25%	SGD
VGG19+LSTM+SVM	83%	82.25%	82%	81.75%	
ResNet50+LSTM+SVM	87%	86.25%	85.75%	86%	
ResNet152+LSTM+SVM	77%	75%	75%	74.5%	
EfficientNetB3+LSTM+SVM	75%	73.25%	73.25%	73%	
VGG16+LSTM+SVM	83%	82.5%	82.25%	82.25%	AdaGrad
VGG19+LSTM+SVM	84%	82.75%	82.75%	82.75%	
ResNet50+LSTM+SVM	88%	88%	87.75%	88%	
ResNet152+LSTM+SVM	81%	80.25%	80%	79.75%	
EfficientNetB3+LSTM+SVM	76%	74.75%	74.5%	74.25%	
VGG16+LSTM+SVM	94%	94%	93.5%	93.5%	RMSprop
VGG19+LSTM+SVM	96%	95.5%	95.25%	95.25%	
ResNet50+LSTM+SVM	99%	99%	98.75%	98.75%	
ResNet152+LSTM+SVM	99%	99%	99%	99%	
EfficientNetB3+LSTM+SVM	98%	98%	97.75%	98%	

accuracies of 98% and 97–98%, respectively. This consistent high performance underscores Adam's effectiveness in managing complex loss landscapes and providing stable convergence for diverse and deep network architectures. In stark contrast, the SGD optimizer led to a substantial performance degradation, with accuracies falling to a range of 75% to 87%. A notable finding here is the variable impact on different architectures. While ResNet50 remained the most robust, achieving 87% accuracy, the performance of the deeper ResNet152 and EfficientNetB3 models collapsed to 77% and 75%, respectively. This suggests that SGD's fixed learning rate and lack of adaptive momentum are particularly detrimental to optimizing the most complex models in the hybrid framework. The AdaGrad optimizer yielded only a marginal improvement over SGD, with accuracy values ranging from 76 to 88%. Once again, ResNet50 demonstrated relative resilience, achieving the highest accuracy in this group at 88%. However, the performance of ResNet152 (81%) and EfficientNetB3 (76%) remained severely limited. This pattern is likely attributable to AdaGrad's aggressive, cumulative learning rate decay, which prematurely reduces update steps and hinders effective convergence in deeper networks. Conversely, the RMSprop optimizer produced results that were highly competitive with, and in the case of ResNet152, identical to those of Adam. It achieved 99% accuracy for both ResNet50 and ResNet152, with the latter also achieving a perfect balance across precision, recall, and F1 score (99% each). VGG19 (96%) and EfficientNetB3 (98%) also performed strongly, confirming RMSprop's capability for delivering stable and effective weight updates, making it a powerful alternative to Adam. In conclusion, the analysis unequivocally demonstrates that the selection of an optimizer is paramount for the success of hybrid CNN-LSTM-SVM models. Adaptive optimizers like Adam and RMSprop are essential for unlocking the full potential of these architectures, particularly for deeper networks such as ResNet152 and EfficientNetB3. The significant performance deficit observed with SGD and AdaGrad highlights their inadequacy for this complex task. Therefore, optimal results are contingent upon a synergistic pairing of a powerful network architecture with a sophisticated, adaptive optimization algorithm.

4.4 Discussion

The rapid advancement of deep learning techniques has substantially transformed the landscape of brain tumor classification from MRI images. Despite significant progress, achieving high accuracy and clinical reliability remains challenging due to the inherent heterogeneity of tumor types, sizes, shapes, and locations, as well as variations in MRI acquisition protocols. Traditional CNN-based and transformer-based models have achieved promising results; however, limitations such as sensitivity to intensity variations, limited ability to model sequential dependencies, and high computational costs continue to constrain their generalizability and practical deployment. In this context, hybrid architectures that combine complementary strengths from multiple deep learning paradigms offer a promising solution for improving both classification performance and robustness. Table 3 provides a comparative overview of these approaches, highlighting the advantages of the proposed CNN + LSTM + SVM hybrid framework in addressing these challenges. Table 4 provides a comprehensive comparison between the proposed hybrid ResNet + LSTM + SVM architectures and several recent state-of-the-art approaches for brain tumor classification using MRI images. The comparison highlights significant progress in classification accuracy, precision, recall, and F1 score achieved by advanced deep learning frameworks.

In the study by Asif et al. (2023b), several deep learning models were enhanced with the Deep Dense Block (DDB) mechanism to improve feature propagation and gradient flow across layers. Among their proposed variants, ResNet152V2+DDB achieved the highest accuracy of 96.10%, followed by Xception+DDB (95.61%) and DenseNet201+DDB (95.12%). These results demonstrate the effectiveness of DDB in addressing the vanishing gradient problem and enriching the discriminative capability of deep architectures. However, despite their strong performance, these models remain sensitive to variations in MRI intensity and tumor heterogeneity, limiting their generalization across datasets.

Khaliki and Başarslan (2024) evaluated multiple transfer learning models, including VGG16, VGG19, InceptionV3, and EfficientNetB4. Their results show that traditional convolutional architectures like VGG16 still achieve remarkable accuracy (98%), owing to their stable feature extraction capabilities and consistent convergence behavior. While deeper and more complex architectures such as InceptionV3 and EfficientNetB4 also attained high accuracy (96–97%), their marginal performance gain did not justify the increased computational cost, indicating that well-tuned conventional CNNs can remain competitive for specific medical imaging tasks.

In contrast, Bilal and Asif (2025) proposed MobileNet-based architectures focusing on lightweight and computationally efficient designs. Their feature fusion model (without attention) reached an accuracy of 96.08%, outperforming the standalone MobileNet-V1 (94.13%) and MobileNet-V2 (92.49%) models. This improvement suggests that multi-feature fusion can effectively integrate low-level spatial and high-level semantic features, improving tumor localization and boundary distinction in

Table 4 Comparative performance with state-of-the-art methods

Reference	Model	Accuracy	Precision	Recall	F1 score
(Asif et al. 2023b)	DenseNet201 + deep dense block (DDB)	95.12%	94.36%	95.05%	95.10%
	Xception + DDB	95.61%	94.98%	95.41%	95.63%
	DenseNet121 + DDB	93.98%	92.87%	93.76%	93.33%
	InceptionResNetV2 + DDB	92.68%	91.94%	91.57%	92.02%
	ResNet152V2 + DDB	96.10%	95.23%	96.01%	96.65%
(Khaliki and Başarslan 2024)	VGG16	98%	98%	98%	97%
	VGG19	96%	96%	96%	96%
	InceptionV3	96%	96%	96%	96%
	EfficientNETB4	97%	97%	97%	96%
	3-layer CNN	91%	91%	91%	90%
(Bilal and Asif 2025)	MobileNet-V1 (Base 1)	94.13%	92.89%	93.73%	93.31%
	MobileNet-V2 (Base 2)	92.49%	91.92%	90.87%	91.39%
	Feature fusion (w/o attention)	96.08%	96.88%	96.38%	96.63%
(Kantu et al. 2025)	SqueezeNet	96.64%	97%	97%	97%
(Amin et al. 2025)	VGG16	95.31%	95.68%	95.37%	95.53%
	VGG19	94.92%	95.48%	95.13%	95.30%
	MobileNetV3	94.92%	95.16%	94.88%	95.02%
	ResNet50	95.70%	96.08%	95.60%	95.84%
	NasNet	94.53%	94.77%	94.48%	94.63%
	ADE_DieT	96.09%	96.10%	96.09%	96.09%
	DeiT	94.92%	95.92%	95.08%	95.50%
(Gupta et al. 2025)	Resnet50	97.86%	97.76%	97.70%	97.72%
	Denesenet121	97.25%	97.16%	97.03%	97.09%
	VGG16	97.56%	97.44%	97.36%	97.39%
	Features concatenation-based brain tumor classification (FCBTC) model-1 (VGG16 + ResNet50)	98.02%	97.96%	97.85%	97.90%
	FCBTC Model-2 (DeneseNet121 + VGG16)	98.25%	98.21%	98.10%	98.15%
	FCBTC Model-3 (ResNet50 + DeneseNet121)	98.40%	98.33%	98.26%	98.27%
This work	ResNet50 + LSTM + SVM (Adam)	99%	99%	98.75%	98.75%
	ResNet152 + LSTM + SVM (Adam)	99%	98.75%	99%	99%
	ResNet50 + LSTM + SVM (RMSprop)	99%	99%	98.75%	98.75%
	ResNet152 + LSTM + SVM (RMSprop)	99%	99%	99%	99%

MRI images. Kantu et al. (2025) explored the potential of SqueezeNet, a compact CNN architecture, achieving an accuracy of 96.64%. The results confirm that lightweight models, when properly optimized, can deliver competitive accuracy with significantly reduced computational complexity, which is advantageous for deployment in real-time or resource-constrained clinical environments. Further performance enhancement was observed in Amin et al. (2025), who conducted an extensive comparative study involving classical CNNs (VGG16, VGG19, ResNet50, MobileNetV3, and NasNet) and vision transformer architectures (DeiT and ADE_DieT). Their ADE_DieT model achieved an accuracy of 96.09%, demonstrating that transformer-based models can effectively capture global dependencies within MRI data. However, their complexity and data requirements remain challenges for small medical datasets.

A remarkable advancement was achieved by Gupta et al. (2025), who proposed feature concatenation-based brain tumor classification (FCBTC) models that combined deep features extracted from multiple pre-trained CNNs (e.g., VGG16 + ResNet50, DenseNet121 + VGG16, and ResNet50 + DenseNet121). Among them, FCBTC Model-3 (ResNet50 + DenseNet121) achieved the highest accuracy of 98.40%, precision of 98.33%, recall of 98.26%, and F1 score of 98.27%. This multi-model fusion approach highlights the benefits of leveraging complementary features across architectures to capture diverse tumor patterns and intensity variations.

In comparison, the proposed hybrid ResNet + LSTM + SVM framework substantially outperforms the aforementioned methods. By combining ResNet for spatial feature extraction, LSTM for sequential feature modeling, and SVM for high-margin classification, the model achieves up

to 99% accuracy, precision, recall, and F1 score using both Adam and RMSprop optimizers. These consistently high scores indicate stability, adaptability, and resistance to overfitting. Unlike purely convolutional or transformer-based models, the inclusion of LSTM allows the network to capture temporal and spatial dependencies across MRI slices, while the SVM enhances decision boundaries in the feature space, resulting in superior discriminative performance.

The analysis of our results reveals several key insights. Positively, the proposed ResNet + LSTM + SVM hybrid architectures consistently achieve high classification accuracy, particularly for no-tumor and pituitary categories, and exhibit robustness across Adam and RMSprop optimizers. Compared to existing methods such as DenseNet+DDB (Asif et al. 2023b), MobileNet fusion (Bilal and Asif 2025), and FCBTC models (Gupta et al. 2025), our approach demonstrates superior performance (up to 99% accuracy), indicating the effectiveness of combining spatial feature extraction, sequential modeling, and SVM classification. On the other hand, challenges remain: misclassifications primarily occur between glioma and meningioma, reflecting intrinsic similarities in MRI appearances. Additionally, model performance is sensitive to optimizer choice, with SGD and AdaGrad yielding substantially lower accuracies for deeper architectures. These findings emphasize the importance of both architecture selection and optimization strategy. Overall, the results underline the potential of hybrid architectures to enhance discriminative power and generalization, while suggesting avenues for further improvement, including the incorporation of attention mechanisms, expansion of datasets, and integration of multi-modal imaging data.

5 Conclusion

Accurate classification of brain tumors from MRI images is a critical task in medical diagnostics, as it directly affects treatment planning, prognosis, and patient outcomes. In this study, we proposed a hybrid framework integrating Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and Support Vector Machines (SVMs) to address challenges arising from tumor heterogeneity and variability in MRI data. Among the evaluated models, the ResNet152+LSTM+SVM architecture achieved up to 99% accuracy, precision, recall, and F1 score, demonstrating superior performance compared to existing state-of-the-art methods.

From this work, several key insights emerge. First, hybrid models that combine deep learning and classical machine learning improve robustness by capturing both local and global tumor features. Second, sequential spatial modeling with LSTM networks enhances the ability to capture long-range dependencies within MRI feature maps. Third, the use of SVM as a classifier provides precise decision boundaries in high-dimensional spaces, complementing the representational strength of CNNs. Finally, interpretability alongside accuracy is crucial for clinical adoption, as hybrid models provide results that can be understood and trusted by medical professionals.

Future research directions include validating the proposed framework on larger, multi-institutional datasets to improve generalizability across diverse MRI scanners and acquisition protocols. The integration of Explainable AI (XAI) techniques, such as Grad-CAM or Score-CAM, will further enhance model interpretability. Optimizing the hybrid architecture through model compression, pruning, or quantization could reduce computational complexity, enabling real-time deployment in clinical settings. Additionally, incorporating multimodal data including CT, PET, or clinical metadata may further enhance classification performance and robustness.

The overarching vision of this work is to develop a fully automated, accurate, and interpretable computer-aided diagnosis (CAD) system that can seamlessly assist radiologists in early detection and personalized treatment planning. By leveraging the complementary strengths of CNNs, LSTM, and SVM, this study lays the foundation for future AI tools that are both clinically effective and operationally feasible in neuro-oncology.

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Data availability The dataset used in this study is public and can be found at the following links: https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset

Declarations

Consent for publication Not applicable.

Competing interests No conflict of interest to report.

Ethics approval and consent to participate Not applicable.

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